NICO BURKHOLDER

Backend & AI Developer | Berlin, Germany (open to remote)
nico.burkholder@web.de github.com/debaser36 unhalteproblem.de

SUMMARY

Backend developer and Computer Science student (TU Berlin, B.Sc. expected 2027). I build typed, tested TypeScript and PostgreSQL services with authentication, API design and deployment, and I am now focusing on agentic coding and AI model evaluation.

EXPERIENCE

DataAnnotation.Tech **Remote**
AI Data Annotation (Contract) *Sep 2026 – present*
- Review and evaluate AI model outputs for accuracy and quality.

Independent study and travel **Mar 2026 – Aug 2026**
- Travelled and taught myself agentic software development; built the full-stack website unhalteproblem.de (see Projects).

Technische Universität Berlin **Berlin, Germany**
Software Developer *Jun 2024 – Feb 2026*
- Built and ran, in a two-person backend team, a web platform for learning formal proofs. It is used for graded homework in two courses (Formale Sprachen und Automaten; Einführung in die theoretische Informatik) by 300–600 students per semester.
- Designed a modular, extensible backend in Effect (a typed TypeScript framework for errors and dependency injection) with a PostgreSQL persistence layer and an OpenAPI contract used by the frontend.
- Implemented login with OpenID Connect (Keycloak), server-side sessions and role-based authorization.
- Connected the backend to Isabelle, a machine-checked proof assistant, for background processing.
- Containerised with Docker, covered by automated tests, and shipped through CI/CD to Linux servers.

Knippenbergs **Berlin, Germany**
Market Sales & Warehouse (part-time) *Dec 2019 – Oct 2022, Jul 2025 – Feb 2026*
- Sold speciality cheese at weekly markets and worked in the warehouse; looked after regular customers ("Stammkunden").

PROJECTS

unhalteproblem.de | personal website and backend **2026**
- Full-stack site with an Effect backend (CQRS, PostgreSQL), Keycloak login with role-based access, a React frontend and a reproducible Nix dev environment.

Own NixOS configuration | github.com/debaser36/nixosConfig **2025**
- Reproducible NixOS setup with flakes and modular Nix modules for four hosts, sops-nix encrypted secrets and a CI flake check.

EDUCATION

Technische Universität Berlin | B.Sc. Computer Science **Expected 2027**
- Coursework: Computer Architecture, Database Management, Cybersecurity, Theoretical Computer Science, Reactive Systems

Herwegh-Gymnasium, Berlin | Abitur **2019**

SKILLS

Backend TypeScript, JavaScript, Node.js, Effect, PostgreSQL, REST/OpenAPI, OpenID Connect (Keycloak)

DevOps Docker, Linux, CI/CD, Git, NixOS

Also Python, C, C++, Haskell, Isabelle, Nix, LaTeX

Languages German (native), English (fluent)